

Assignment 1 - Conceptualizing an agentic autonomous GIS framework

Submission Date: Sunday, September 13th, 2026

Think about how you can apply an Agentic GeoAI framework (e.g., an autonomous GIS framework) to your own academic or professional research interests. You can begin by thinking about the tasks you have carried out manually that could subsequently be automated by an agent.

Create a 6-minute presentation where you introduce your background, formulate a meaningful geographic research problem, and propose a small agentic (autonomous GIS) application that could be developed further in this class, for example, as the basis of your final coursework project.

The emphasis is on thoughtful problem formulation and a possible design approach, not on building a finished application. Moreover, grading will not be strict on architectural considerations, as we have not covered that in class yet (this assignment is more about developing a concept/idea).

Your proposal should make clear which decisions belong to the human researcher and which tasks could reasonably be delegated to an autonomous agent. Also consider how you may need to determine whether the resulting analysis is correct and defensible.

Table 1 outlines the important framework dimensions to consider.

Framework component	Question to address
Objective	What specific outcome should the agent produce?
User context	Who will use the system and expected output? What decision or research activity will it support?
Inputs	Which spatial and non-spatial data are required?
Workflow	What might be a possible conceptual sequence of automation for your chosen problem?
Tools	Which GIS operations, code libraries, databases, APIs, models, or other tools may be utilized?
Human-agent responsibilities	Which tasks may be delegated? What decisions must remain with a human?
Outputs	What artifacts will be created, and what conclusions may be drawn from them?
Validation	What automated checks would provide evidence that the result is correct?
Failure modes	How could the workflow fail through bad data, hallucinations, bias, or other issues? How will these risks be limited?
Scope and cost	How will you keep the prototype feasible with economical use of model tokens?

Table 1 Framework aspects to consider.

Please submit your presentation online as a pdf. You will then present your pitch in the next class.

Table 2 then outlines the assessment criteria to consider.

Finally, students may use AI tools in accordance with the course AI policy. Yet, students remain responsible for understanding, testing, validating, and documenting all submitted material. The presentation must include a brief AI-use statement identifying the tools used and explaining how they contributed. Undisclosed AI-generated material, fabricated citations and other materials, or code that the student cannot explain may constitute an Honor Code violation and will be reported.

Criterion	Evidence	Weight
Personal and course connection	The presentation clearly links the student's background and learning goals to the proposed work.	20%
Research problem and significance	The topic is geographically meaningful and identifies relevant users or stakeholders.	20%
Agentic autonomous framework	The proposed workflow clearly defines inputs, tools, agent actions, human responsibilities and outputs.	20%
Validation	The proposal identifies limitations and offers credible validation checks (to ensure correct answers).	20%
Communication	Slides are clear and legible, sources are acknowledged, and the presentation stays within the assigned time and scope.	20%

Table 2 Assessment criteria.